

Shweta Smriti Tripathi

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EDUCATION

Columbia University (GPA: 3.6/4.0)

Master of Science in Data Science

- Coursework: Agentic AI, GenAI with LLMs, Applied Deep Learning, Statistical Inference, HPML, EDAY.

New York, NY

Aug 2025 – Dec 2026

Kalinga Institute of Industrial Technology (GPA: 3.99/4.0)

B.Tech in Computer Science and Engineering

- Coursework: Data Structures, DBMS, Software Engineering, Machine Learning, Artificial Intelligence.

Odisha, India

Jun 2018 – Jul 2022

SKILLS

Languages: Python, SQL, Bash, Java, R

Data Engineering: Spark, PySpark, Databricks, Airflow, dbt, Kafka, ETL/ELT, Batch and Streaming Pipelines, A/B Testing

Data Modeling and Warehousing: Dimensional Modeling, Schema Design, Snowflake, Teradata, BigQuery, Azure SQL, S3

Data Quality and Governance: Data Contracts, Schema Validation, Data Quality, Data Validation, Pipeline Monitoring, CI/CD

Cloud and APIs: AWS, Azure, Docker, Kubernetes, Git, REST APIs, FastAPI, MCP (Model Context Protocol)

Analytics and ML: Power BI, Tableau, Pandas, NumPy, scikit-learn, PyTorch, TensorFlow, XGBoost, LightGBM

EXPERIENCE

Amazon

Seattle, WA

Data Engineer Intern

May 2026 – Aug 2026

- Automated 10+ recurring data engineering tasks by developing an MCP server from daily workflow analysis, reducing manual engineering effort by 40%.
- Expanded the MCP server into a reusable framework within the internal AI platform, enabling 3+ teams to automate data requests and reducing turnaround time by 30%.
- Established a production metrics pipeline for Weekly Business Reviews by processing 50K+ seller interactions, powering leadership dashboards for response and engagement analysis.

Providence Global Center (Client: AdventHealth)

Hyderabad, India

Senior Data Engineer

Jul 2024 – Jul 2025

- Accelerated TB-scale data refreshes from 8 hours to 30 minutes with Airflow and Databricks by optimizing partitioned loads and Snowflake clustering, supporting 70+ users across 6 teams.
- Modeled 10M+ daily security events from Kafka and S3 in Snowflake using dimensional schemas, supporting 45+ KPIs and reducing compute spend by 35% for CISO reporting.
- Strengthened data quality with dbt contracts, snapshots, CI/CD checks, schema-drift alerts, and freshness SLOs, raising on-time pipeline SLA from 91% to 99%.
- Coordinated requirements with security and business stakeholders, translating reporting needs into production datasets while supporting incident runbooks and postmortems across 19 HITRUST domains.

Dell Technologies

Bangalore, India

Business Intelligence and Data Engineer

Jun 2022 – Jul 2024

- Orchestrated 80+ daily Airflow jobs with incremental loads and automated data checks, reducing manual runbook effort by 75% and improving reporting availability for Sales and Operations.
- Integrated Salesforce API data with 4M+ Teradata records using PySpark and SQL ETL, shortening processing windows by 60% and stabilizing daily GTM reporting.
- Standardized KPIs across 30+ Power BI dashboards for EMEA and LATAM by aligning definitions with Sales Ops stakeholders, reducing metric discrepancy escalations by 55%.

PROJECTS

SQL Copilot Agent: Text to SQL with Guardrails | FastAPI, LangGraph, GPT-4, Claude, SQL, Snowflake

- Created a Snowflake text-to-SQL system that maps business questions to schema-grounded queries, achieving 93% execution success across 120+ prompts with 3.9s median latency.
- Controlled query execution with read-only access, schema and join validation, and structured outputs, reducing invalid SQL from 28% to 6%.

MacroCart: Data and Inference Pipeline | Python, SQL, FastAPI, Chroma, OpenAI, Ollama

- Delivered a structured nutrition planning platform that generated multi-meal plans under daily macro constraints across 120+ test scenarios.
- Validated serving data through typed FastAPI models and Chroma retrieval, reducing missing and inconsistent records by 45% and raising validation pass rates from 82% to 96%.

Home Credit Default Pipeline | SQL, Feature Engineering, Reproducible Transforms

- Designed leakage-safe transformations and automated feature engineering for 300K+ applications, improving PR-AUC from 0.46 to 0.53 and AUC from 0.76 to 0.79.